

Assignment No: 2(B)

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#include <iostream>
#include <omp.h>

using namespace std;

void merge(int arr[], int l, int m, int r)
{
    int n1 = m - l + 1;
    int n2 = r - m;

    int L[n1], R[n2];

    for(int i = 0; i < n1; i++)
        L[i] = arr[l + i];

    for(int j = 0; j < n2; j++)
        R[j] = arr[m + 1 + j];

    int i = 0, j = 0, k = l;

    while(i < n1 && j < n2)
    {
        if(L[i] <= R[j])
            arr[k++] = L[i++];
        else
            arr[k++] = R[j++];
    }

    while(i < n1)
        arr[k++] = L[i++];

    while(j < n2)
        arr[k++] = R[j++];
}

void sequentialMergeSort(int arr[], int l, int r)
{
    if(l < r)
    {
        int m = (l + r) / 2;

        sequentialMergeSort(arr, l, m);
        sequentialMergeSort(arr, m + 1, r);
    }
}
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        merge(arr, l, m, r);
    }
}

void parallelMergeSort(int arr[], int l, int r)
{
    if(l < r)
    {
        int m = (l + r) / 2;

        #pragma omp parallel sections
        {
            #pragma omp section
            parallelMergeSort(arr, l, m);

            #pragma omp section
            parallelMergeSort(arr, m + 1, r);
        }

        merge(arr, l, m, r);
    }
}

void printArray(int arr[], int n)
{
    for(int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main()
{
    int n;

    cout << "Enter number of elements: ";
    cin >> n;

    int arr[n], arr2[n];

    cout << "Enter elements:\n";

    for(int i = 0; i < n; i++)
    {
        cin >> arr[i];
        arr2[i] = arr[i];
    }
}

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double start, end;

// Sequential Merge Sort
start = omp_get_wtime();
sequentialMergeSort(arr, 0, n - 1);
end = omp_get_wtime();

cout << "\nSequential Sorted Array:\n";
printArray(arr, n);
cout << "Sequential Time: " << end - start << " seconds\n";

// Parallel Merge Sort
start = omp_get_wtime();
parallelMergeSort(arr2, 0, n - 1);
end = omp_get_wtime();

cout << "\nParallel Sorted Array:\n";
printArray(arr2, n);
cout << "Parallel Time: " << end - start << " seconds\n";

return 0;
}

```

Input:

Enter number of elements: 6

Enter elements:

9 4 7 1 3 6

Output:

Sequential Sorted Array:

1 3 4 6 7 9

Sequential Time: 0.00003 seconds

Parallel Sorted Array:

1 3 4 6 7 9

Parallel Time: 0.00001 seconds